Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[18BS5AI01]**

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

STATISTICS USING R

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Distinguish between inferential statistics and descriptive statistics. **07**

b. Construct Histogram and list out the minimum five analysis which you observe from the following distribution. **07**

Monthly Wages (in Rs.)	0-1000	1000-2000	2000-3000	3000-4000	4000-5000	5000-6000	6000-7000	7000-8000
No. of family	21	35	56	74	63	40	29	14

2. a. Determine the quartile deviation and the coefficient of quartile deviation for the following data **06**

x	351 – 500	501 – 650	651 – 800	801–950	951–1100
f	48	189	88	47	28

b. From the data given below, calculate the first four moments about an arbitrary origin and then calculate the first four moments about the mean and also calculate $\beta 1$ and $\beta 2$ for the following data. **08**

x	30-33	33-36	36-39	39-42	42-45	45-48
f	2	4	26	47	15	6

3. a. Show that Fisher's index number satisfies both time reversal and factor reversal tests. **07**

b. Compute the chain index number with 2003 prices as base from the following table giving the average whole sale prices of commodities A, B, C for the year 2003 to 2007. **07**

Commodity	Average whole sale price (in Rs.)				
	2003	2004	2005	2006	2007
A	20	16	28	35	21
B	25	30	24	36	45
C	20	25	30	24	30

OR

4. a. Construct index number of price from the following data by applying **07**

- **i.** Laspeyre' s method
- **ii.** Paasche' s method
- **iii.** Fisher' s ideal index method.

	2000		2001	
Commodity	Price	Quantity	Price	Quantity
A	2	8	4	5
B	5	12	6	10
C	4	15	5	12
D	2	18	4	20

b. Calculate consumer price index by using Family Budget method for year 1993 with 1990 as base year from the following data. **07**

	Price		
Items	1990 (Rs.)	1993 (Rs.)	Weight
Food	150	140	35
Rent	75	90	20
Clothing	25	30	10
Fuel and lighting	50	60	15
Miscellaneous	60	80	20

5. a. Discuss the mathematical models for time series analysis. **05**

b. The following are the figures of production (in thousand quintals) of a sugar factory. **09**

Year	1992	1994	1996	1998	2000	2002	2004
Production	77	81	88	94	94	96	98

i. Fit a straight line trend to the data.

ii. Graph the observed values and the trend values.

iii. Estimate the production in the year 2006

OR

6. a. Calculate the trend value to the following data by the method of semi- averages. **07**

Year	1995	1996	1997	1998	1999	2000	2001
Expenditure (Rs. in Lakhs)	1.5	1.8	2.0	2.3	2.4	2.6	3.0

b. Determine the seasonal indices from the following data using simple average method. **07**

Year

Quarter	1974	1975	1976	1977	1978
I	72	76	74	76	74
II	68	70	66	74	74
III	80	82	84	84	86
IV	70	74	80	78	82

7. a. Write R commands to generate a matrix $A = \begin{bmatrix} 2 & 7 & 8 \\ 6 & 8 & 10 \end{bmatrix}$ and extract the following **04**

Using R-commands

i. Elements at 2nd row 3rd column.

ii. The entire 2nd row.

iii. The entire 3rd column.

iv. First and third column.

v. Find the transpose of the matrix A.

b. Write R commands to find **05**

i. Relative frequency distribution of the eruption durations in faithful

ii. The histogram of the eruption durations in faithful.

c. Write R commands for the following **05**

i. The log to the base 10 of X1= 73.

ii. The square root of X2 = 137 by raising it to the power 0.5.

iii. For average of X1 and X2.

iv. If a= (8, 12, 5, 9) and b= (2, 4, 6, 8) are vectors, multiply 5 to a and add with b and store in c and raising the power 2.

OR

8. a. Write R program to construct a matrices $A=\begin{bmatrix} 7 & 14 \\ 9 & 6 \\ 10 & 8 \end{bmatrix}$, $B=\begin{bmatrix} 6 \\ 7 \\ 8 \end{bmatrix}$ and $C=[12 \quad 5]$ and then perform the following using R commands **04**

i. Transpose of matrix A

ii. Combine the columns of A and B.

iii. Combine the rows of A and C

b. Write R commands **05**

i. To create a vector called x1 and x2 consisting of the numbers (1,3,5,7) and (2,4,6,8) respectively

ii. Add x1 and x2 store in a vector V

iii. Subtract x1 from x2

iv. Create a new vector called "x3" by multiplying vector x1 by vector x2

v. Create a new vector called "x4" by taking the square root of each member of x3.

vi.

c. Write R commands to find **05**

i. frequency distribution of the painter schools

ii. Pie chart of the painter school and write a R command to color the pie chart.